

Sports Hall Booking System Using ServiceNow

SmartBridge | AICTE | ServiceNow University

Virtual Internship Program – FY26 Cohort 5 -Batch 4.2

Project Objectives

Phase 1: Requirement Analysis & Planning

1.1: Business Objectives

1. Digitize the Sports Hall Booking Process

The primary objective of this project is to digitize the sports hall booking process using the ServiceNow Service Portal. The system replaces manual booking methods with an automated platform that enables users to reserve sports facilities quickly, efficiently, and transparently.

The system aims to:

- Provide an online platform for booking sports halls.
 - Automate reservation and approval workflows.
 - Improve transparency in facility availability.
 - Reduce manual paperwork and booking errors.
-

2. Enhance User Convenience

The project is designed to offer a user-friendly and intuitive booking experience through the ServiceNow Service Portal.

Key objectives include:

- Allow users to view available sports halls and time slots in real time.
 - Simplify the booking process with an easy-to-use interface.
 - Minimize manual booking requests and physical coordination.
 - Provide instant booking confirmation through automated email notifications.
 - Enable users to manage and track their booking status online.
-

3. Improve Operational Efficiency

Automation plays a major role in improving the overall efficiency of sports facility management.

The system aims to:

- Automate booking approvals and confirmation processes.

- Prevent double bookings using real-time availability validation.
 - Reduce administrative workload for sports facility managers.
 - Maintain centralized booking records for monitoring, auditing, and future reference.
 - Improve accuracy and reduce human errors in booking management.
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4. Enable Secure Digital Payments (Future Enhancement)

As a future enhancement, the system can support secure online payment integration for paid sports facilities.

Proposed features include:

- Online payment processing.
 - Automatic fee calculation based on booking duration and facility type.
 - Digital payment receipts and transaction tracking.
 - Reduced dependency on cash handling and manual reconciliation.
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5. Improve Data Visibility and Reporting

The system provides centralized booking data that can be used for analysis and reporting.

Objectives include:

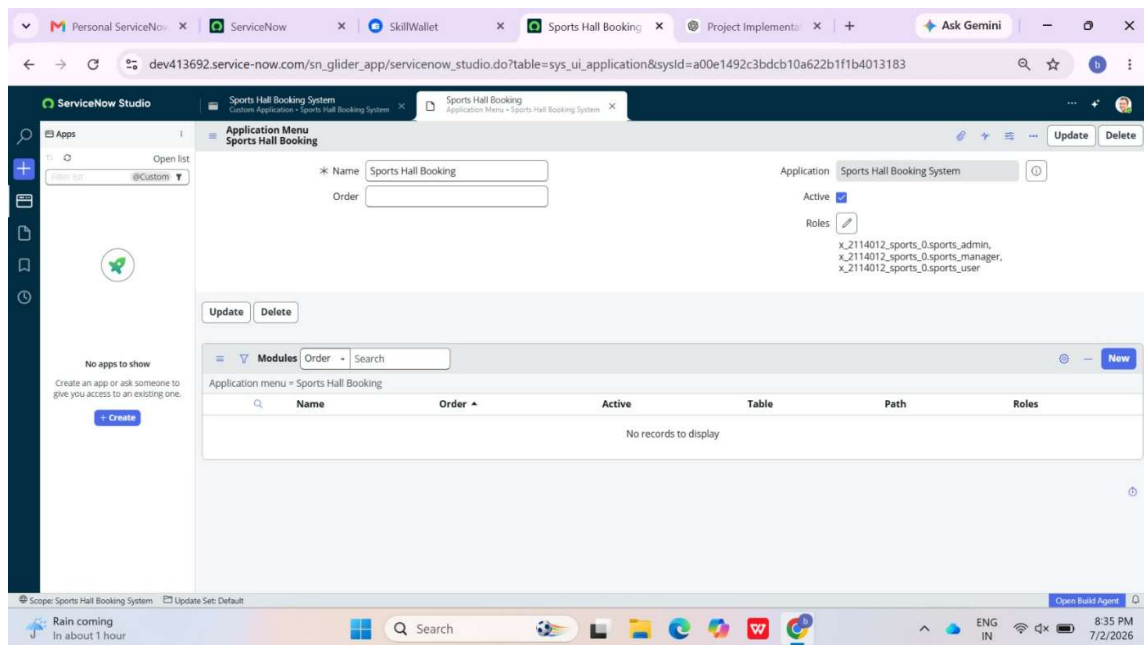
- Track booking trends based on sports type, time slot, and location.
 - Identify peak booking hours and underutilized facilities.
 - Generate reports for administrators.
 - Support data-driven decisions for scheduling, maintenance, and resource allocation.
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6. Promote Resource Optimization and Sustainability

The project encourages efficient utilization of sports infrastructure while supporting environmentally friendly operations.

The system aims to:

- Maximize utilization of available sports halls.
 - Minimize idle facility time.
 - Eliminate paper-based booking registers.
 - Support digital-first operations that reduce administrative overhead and promote sustainability.
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1.2:Functional Scope

1 .Widgets and Pages Configuration

A ServiceNow Widgets and Pages Configuration with the following details:

Sports Hall / Facility Name

Sport Type (Badminton, Basketball, Table Tennis, etc.)

Booking Date

Time Slot (Start Time – End Time)

User Contact Details

2. Booking Validation & Availability

Real-time validation of hall availability

Prevention of overlapping bookings

Configurable booking rules (maximum duration, advance booking limits)

3. Approval & Confirmation Workflow

Optional approval from Sports Hall Manager

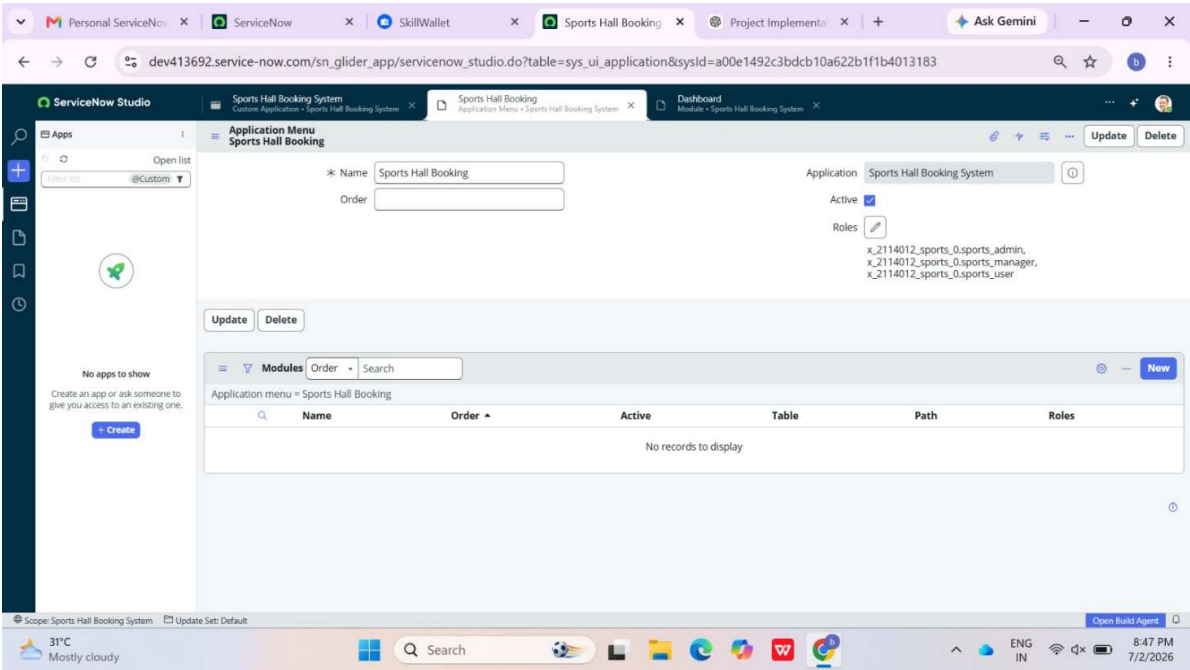
Automated status updates (Pending, Approved, Rejected, Cancelled)

Email notifications to users and administrators

4. Payment & Refund Handling

Automatic fee calculation based on duration and sport

Cancellation charge logic (e.g., 10% deduction),Refund amount calculation and storage

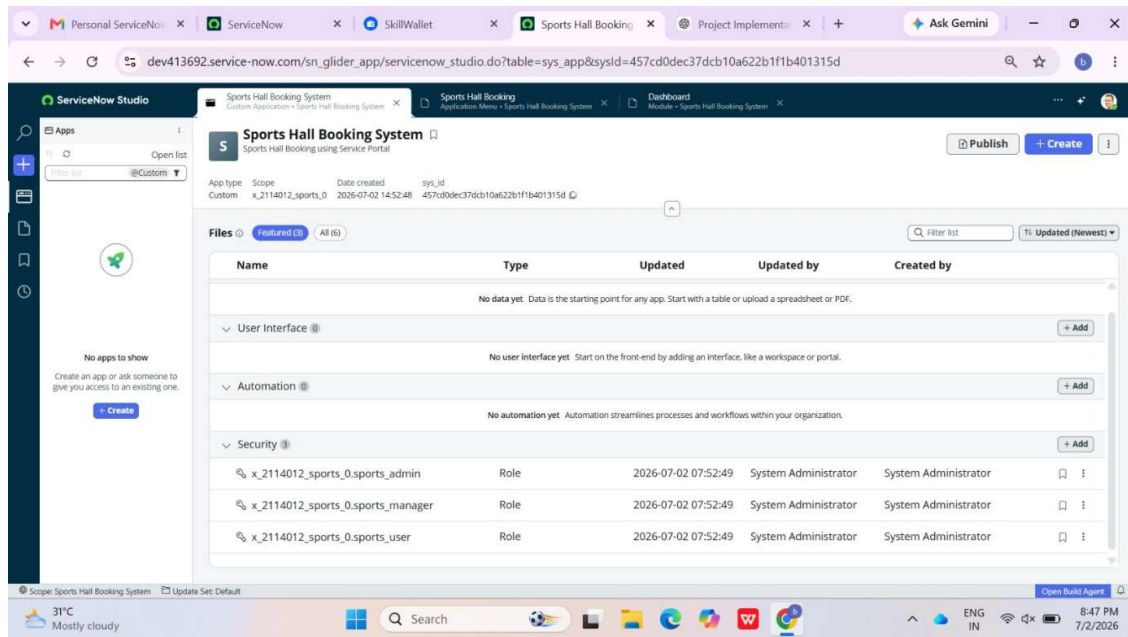


1.3:Stakeholder Mapping

Description

Stakeholder	Responsibility
End Users	Book sports halls and manage reservations
Sports Hall Managers	Approve bookings and manage availability
Facility Administration	Monitor utilization and maintenance

Stakeholder	Responsibility
End Users	Reserve sports halls through the ServiceNow portal, view available time slots, and manage or cancel their bookings.
Sports Hall Managers	Review and approve booking requests, manage hall availability, and ensure efficient scheduling of sports facilities.
Facility Administration	Monitor facility utilization, generate booking reports, oversee maintenance schedules, and manage overall sports infrastructure.



1.4:Execution Roadmap

Key Milestones

Requirement Gathering & Analysis

Service Portal UI Design

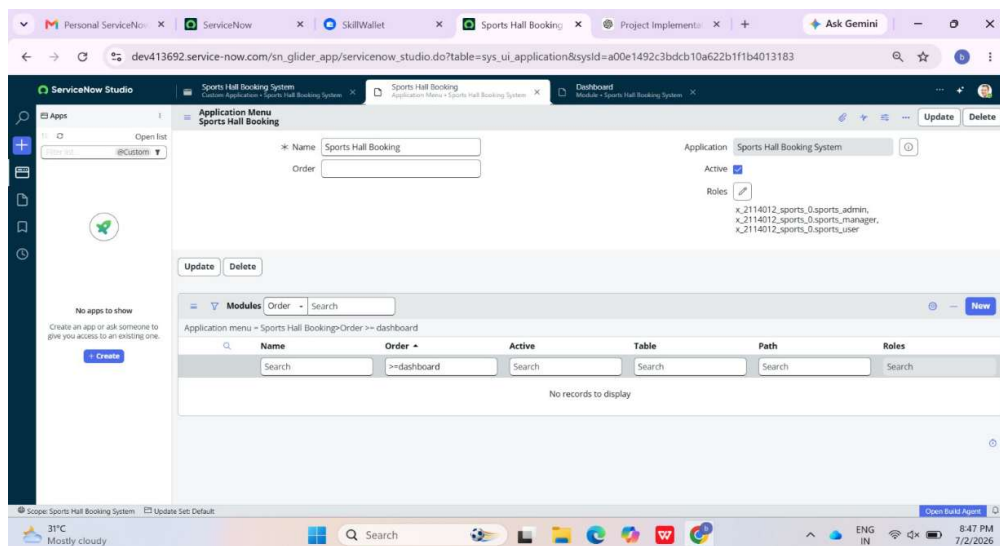
Availability & Booking Logic Implementation

Approval Workflow Setup

Notification & Email Template Design

User Acceptance Testing (UAT)

Production Deployment



Phase 2: Backend Development & Configurations

2.1: Data Architecture:

Creation of Table

Activity 1: Creation of Table

Navigate to: System Definition > Tables.

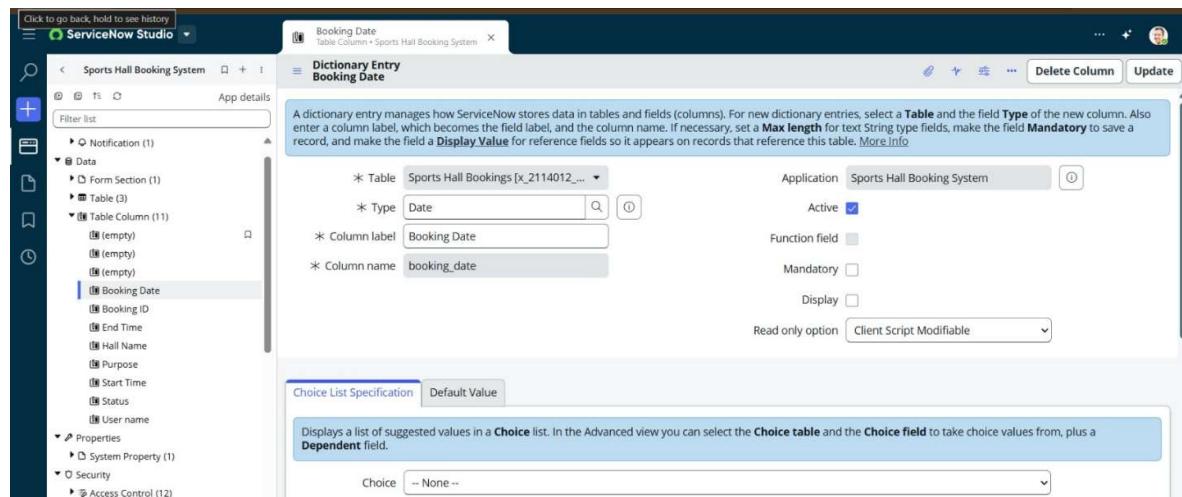
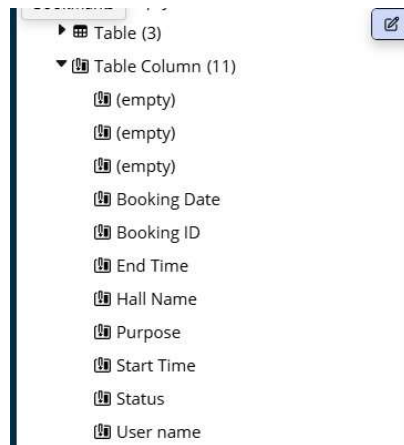
Click New to create a new table.

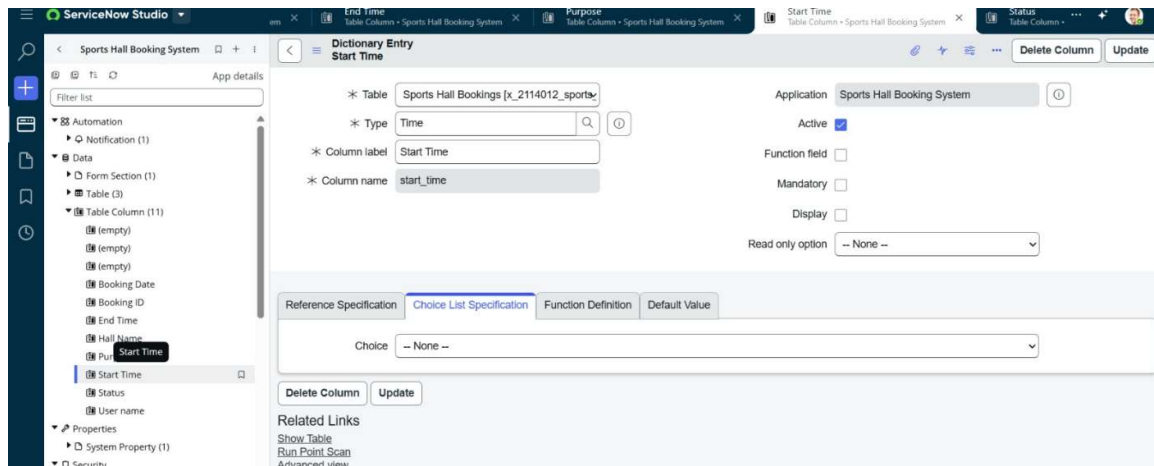
Fill in Table Information:

Name: u_sports_hall_bookings

Label: Sports Hall Bookings

Click Submit to create the table.





Activity 2: Creation of fields

In ServiceNow, fields are created at the table level. To create a field, you first need to identify the table where the field will reside.

1. In the Application Navigator (left-side panel), type Tables in the search bar.
2. Under System Definition, click Tables. This will take you to a list of all tables in the system.

Select the Table to Add the Field

1. From the list of tables, search for and select the table you want to add a field to. For example, if you want to add a field to the Sports Hall Bookings table:
2. Type “Sports Hall Bookings” in the search box or scroll through the list.
3. Click on the Sports Hall Bookings table name. You’ll now see a list of all fields (columns) associated with the Sports Hall Bookings table.

Open the Table’s Columns

1. After selecting the table, you’ll be brought to a view that lists all the columns (fields) that currently exist on that table.
2. To create a new field (column), go to the Columns tab (this is where all fields for the selected table are listed).

Create a New Field

1. In the Columns tab, click the New button located at the top-right corner of the page to create a new field.
2. You’ll now be prompted with a form where you need to define the new field. The following fields need to be filled out:
3. Create the following fields as mentioned in the picture below

Table Sports Hall Bookings						
Columns Controls Application Access						
Table Columns for text Search 1 to 14 of 14 New						
Dictionary Entries						
	Column label	Type	Reference	Max length	Default value	Display
	End Time	Time	(empty)	40	40	false
×	Booking Date	Date	(empty)	40	40	false
	Updates	Integer	(empty)	40	40	false
×	Status	Choice	(empty)	40	40	false
×	Booking ID	String	(empty)	50	50	false
	Updated by	String	(empty)	40	40	false
	Created by	String	(empty)	40	40	false
×	Purpose	String	(empty)	255	255	false
×	Start Time	Time	(empty)	40	40	false
	Created	Date/Time	(empty)	40	40	false
	Updated	Date/Time	(empty)	40	40	false
×	Hall Name	Choice	(empty)	50	50	false
×	User name	Reference	user[sys_user]	32	32	false
	Sys ID	Sys ID (GUID)	(empty)	32	32	false
+	Insert a new row...					

Form Configuration

After creating the table and fields, configure the form layout to provide a user-friendly interface for entering booking details.

Steps:

1. Navigate to **System Definition → Tables**.
2. Open the **Sports Hall Bookings** table.
3. Click **Configure → Form Layout**.
4. Add the required fields to the form.

The form includes the following fields:

- Booking ID
- Sports Hall Name
- Sport Type
- Booking Date
- Start Time
- End Time
- User Name
- Email Address
- Phone Number
- Booking Status
- Approval Status
- Payment Status

- Remarks

Arrange the fields in a logical order for better usability and click **Save**.

The screenshot shows the 'Scheduled Script Execution' form in ServiceNow. The form is titled 'Scheduled Script Execution - Sports Hall -10 MinReminder'. It includes fields for Name, Active (checked), Application (Sports Hall Booking System), and Conditional (unchecked). A blue informational box states: 'For scheduled job types that require an entered time, you have the option to enter an associated time zone. If no time zone is selected, the job will run at the entered time in time zone of the user who entered the time. If 'Use System Time Zone' is selected, the entered time will run in the time zone of the instance running the job.' Below this, there are fields for Run (Periodically), Time zone (None), Repeat Interval (Days: 0, Hours: 00, Minutes: 01, Seconds: 00), Advanced (checked), Starting (2026-07-05 07:27:43), and Ending.

Service Catalog Item Creation

Create a Catalog Item

To allow users to book sports halls through the Service Portal, a Service Catalog Item is created.

Steps:

1. Navigate to Service Catalog → Catalog Definitions → Maintain Items.
2. Click New.
3. Select Catalog Item.
4. Enter the following details

The screenshot shows the 'Scheduled Script Execution' form with the script code entered in the 'Script' field. The code is as follows:

```

1 (function() {
2
3   var now = new GlideDateTime();
4   var fiveMinutesLater = new GlideDateTime();
5   fiveMinutesLater.add(5 * 60 * 1000);
6
7   var sevenMinutesLater = new GlideDateTime();
8   sevenMinutesLater.add(7 * 60 * 1000);
9
10  var gr = new GlideRecord('u_sports_hall_bookings');
11
12  gr.addQuery('u_status', 'pending');
13  gr.addQuery('u_reminder_sent', false);
14  gr.addQuery('u_start_time', '>=', fiveMinutesLater.getValue());
15  gr.addQuery('u_start_time', '<=', sevenMinutesLater.getValue());
16
17  gr.query();
18
19  while (gr.next()) {
20
21    var startTime = gr.u_start_time.getDisplayValue();
22    var email = gr.u_email.toString();
23  }
24 }
  
```

Scheduled Jobs

Scheduled Jobs are configured to automate background tasks.

Examples include:

- Remove expired bookings.
- Update hall availability.
- Archive completed booking records.
- Send reminder emails before scheduled bookings.

This reduces manual maintenance efforts.

Name	Active	Class	Updated
Sports Hall - 10 MinReminder	true	Scheduled Script Execution	2026-07-05 07:32:09

Access Control (ACL)

Access Control Lists (ACLs) are implemented to provide secure access to booking data.

User Permissions

- Create booking requests.
- View personal bookings.
- Cancel own bookings.

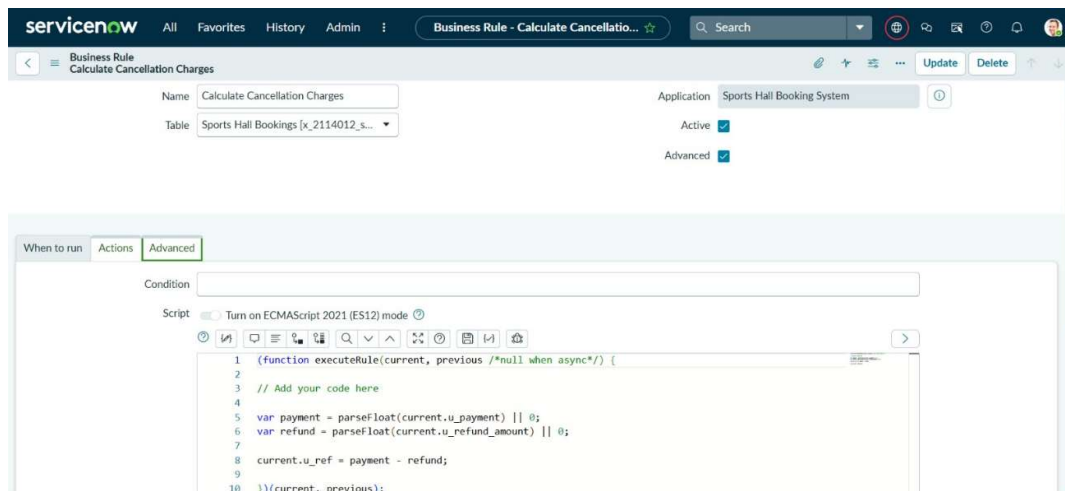
Sports Hall Manager Permissions

- View all booking requests.
- Approve or reject bookings.
- Modify booking status.

Administrator Permissions

- Full access to booking records.
- Configure workflows.
- Manage users.
- Generate reports.

This ensures secure and role-based access throughout the application.



Phase 3: UI/UX Development & Customization

Homepage Design

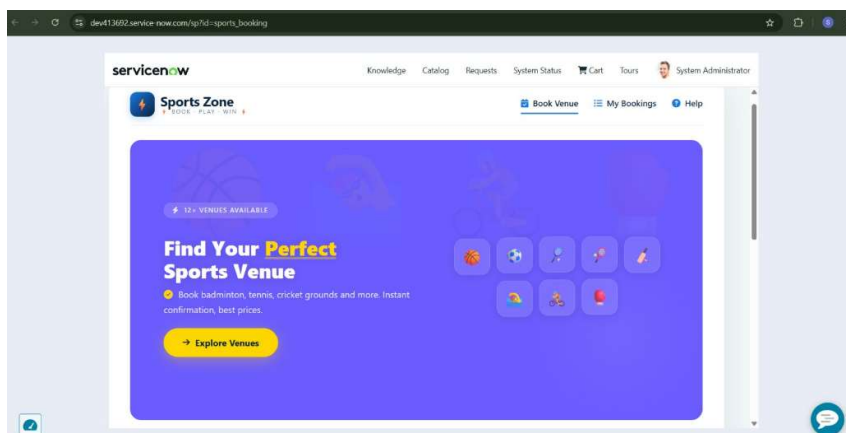
Activity 1: Create Homepage

The homepage acts as the landing page for users visiting the Sports Hall Booking Portal.

The homepage includes:

- Welcome Banner
- Sports Hall Information
- Available Facilities
- Quick Navigation Cards
- Latest Announcements
- Contact Information

The layout was designed using Bootstrap components to ensure a clean and organized appearance.



Design Booking Request Form

The booking form was customized to collect all necessary reservation details.

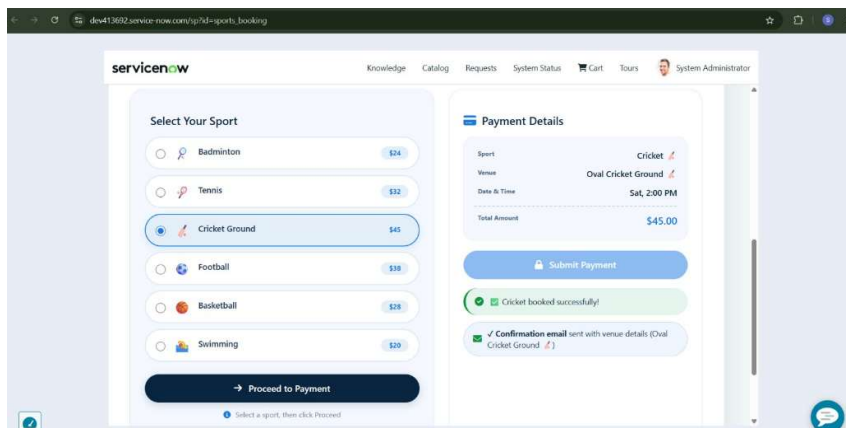
The form includes:

- Sports Hall Selection
- Sport Type
- Booking Date
- Start Time
- End Time
- Contact Number
- Email Address
- Purpose of Booking

Mandatory field validation was enabled to prevent incomplete submissions.

Additional client-side validation ensures:

- Date cannot be in the past.
- End time must be greater than start time.
- Required fields cannot be left empty.



The screenshot shows a web browser window displaying a ServiceNow booking form. The URL in the address bar is `dev413602.servicenow.com/sp?id=sports_booking`. The page has a navigation bar with links: Knowledge, Catalog, Requests, System Status, Cart, Tours, and a user profile icon labeled 'System Administrator'. The main content area is divided into two panels. The left panel, titled 'Select Your Sport', lists several sports with their respective prices: Badminton (\$24), Tennis (\$32), Cricket Ground (\$45), Football (\$18), Basketball (\$28), and Swimming (\$20). The 'Cricket Ground' option is selected. Below the list is a 'Proceed to Payment' button. The right panel, titled 'Payment Details', shows the selected sport as 'Cricket', the venue as 'Oval Cricket Ground', and the date & time as 'Sat, 2:00 PM'. The total amount is \$45.00. There is a 'Submit Payment' button. Below the payment details, there are two status messages: 'Cricket booked successfully!' and 'Confirmation email sent with venue details (Oval Cricket Ground)'. A footer note at the bottom left says 'Select a sport, then click Proceed'.

User Dashboard

A personalized dashboard was created to allow users to monitor all their booking activities.

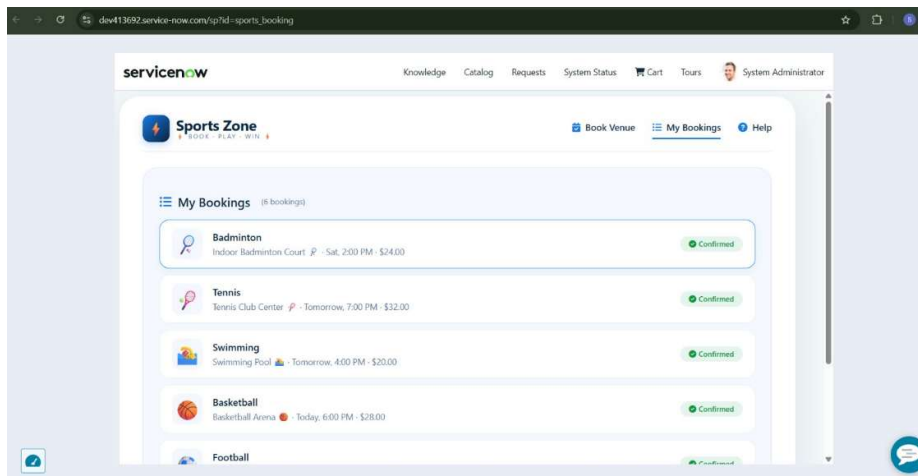
The dashboard displays:

- Total Bookings
- Pending Requests
- Approved Bookings
- Cancelled Bookings

- Upcoming Reservations
- Booking History

Users can also:

- View booking details
- Cancel bookings
- Track approval status



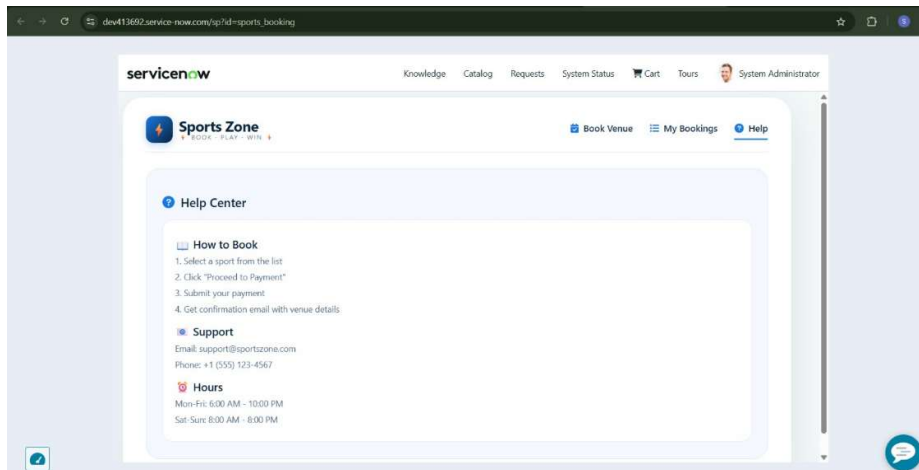
Form Validation

Client Scripts were implemented to perform validations before form submission.

The scripts validate:

- Mandatory fields
- Date selection
- Time slot validation
- Contact number format
- Email format

Users receive instant feedback whenever invalid data is entered.



Phase 4: Data Migration, Testing & Security

Import Initial Data

The first step in Phase 4 was to migrate the required data into the Sports Hall Booking System. Initial records such as sports hall details, facility information, administrator accounts, and sample booking records were imported into the custom table.

Steps:

1. Navigate to System Import Sets → Load Data.
2. Upload the source file (Excel/CSV).
3. Create an Import Set Table.
4. Configure a Transform Map.
5. Map source fields to the target table (Sports Hall Bookings).
6. Execute the transform.
7. Verify the imported records.

Data Migrated

- Sports Hall Details
- Sports Categories
- Time Slots
- Facility Information
- Sample Booking Records
- User Information

The screenshot shows the 'Service Portal - Sports Zone' configuration page in ServiceNow. The page includes fields for Title, URL suffix, Home page, KB home page, Login page, Logo, Logo Alt Text, Icon, Application, 404 page, Catalog home page, Catalog category home page, Main menu, Theme, and Inactive checkbox. The 'CSS variables' section is expanded, showing a list of variables and their values.

Verify Imported Records

After migration, all imported records were validated to ensure accuracy.

The following checks were performed:

- Duplicate record validation
- Missing value verification
- Field mapping validation
- Data consistency check
- Record count verification

Any incorrect records were corrected before proceeding to testing.

Title	URL suffix	Theme	Updated
CAB Workbench	cab	CAB Workbench - Default	2026-04-30 01:05:03
Mobile Employee Service Portal	mesp	Mobile Employee Next Experience Theme	2022-08-08 02:10:34
Knowledge Portal	kb	Customer Experience Coral Theme	2025-02-27 23:20:17
Benchmarks	benchmarks	Benchmarks - Theme	2026-04-30 01:05:03
Employee Center	esc	Employee Center Coral Theme	2026-04-30 00:58:52
Performance Analyzer	perf	(empty)	2026-04-30 01:05:03
Service Portal	sp	Coral	2026-04-30 01:05:03
SP Configuration	sp_config	La Jolla	2026-04-30 01:05:03
Sports Zone	sz	La Jolla	2026-07-05 21:22:47
Service Workspace Portal	swp	Workspace EC Theme	2022-09-14 00:45:58

End User Testing

The application was tested from the user's perspective.

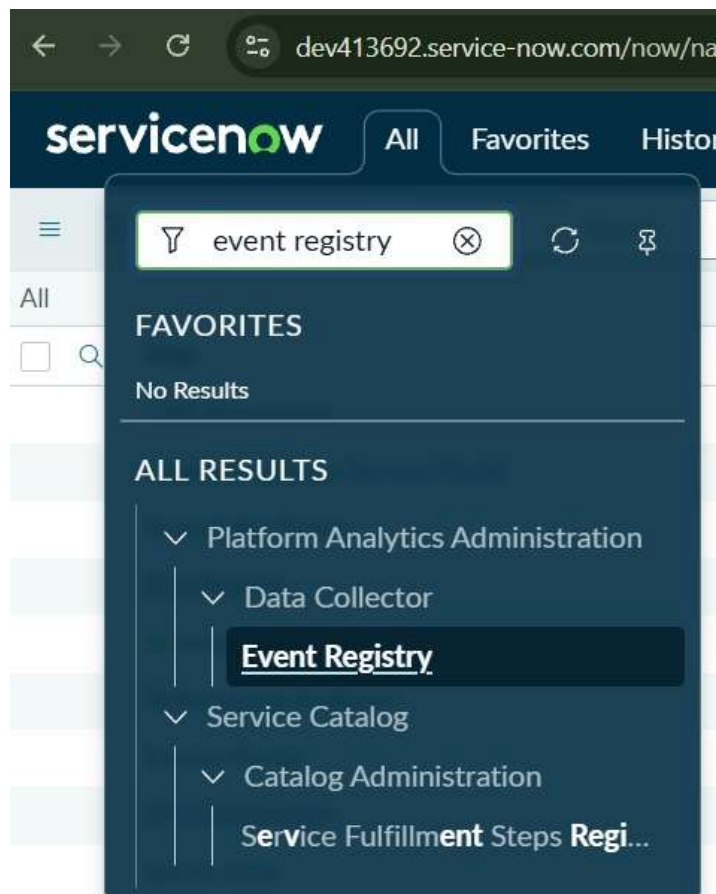
Users performed the following activities:

- Login to Service Portal

- Submit Booking Request
- View Booking Status
- Cancel Booking
- Receive Email Notification
- Track Booking History

Feedback received from users included:

- Easy navigation
- Simple booking process
- Fast approval workflow
- Clear booking status updates
- Responsive user interface



System Performance Evaluation

Performance testing was conducted to verify application responsiveness.

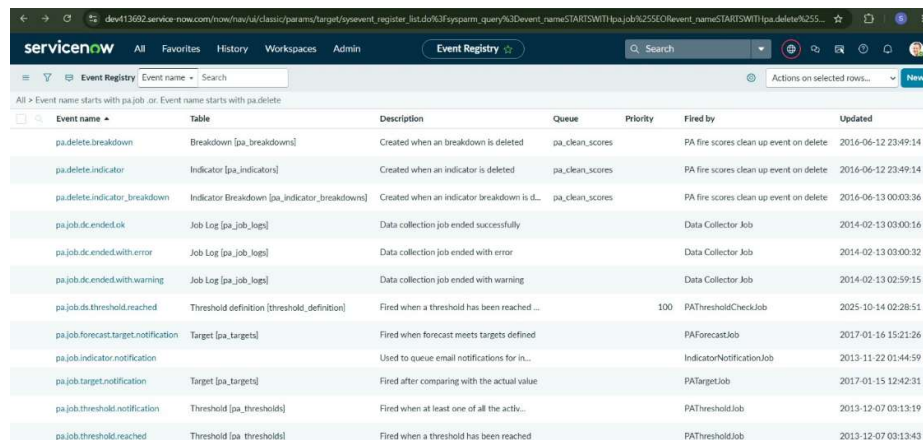
The following parameters were evaluated:

- Page Loading Time

- Form Submission Time
- Booking Processing Time
- Workflow Execution Time
- Dashboard Loading Time

Multiple booking requests were submitted simultaneously to evaluate system performance.

The application maintained stable performance under normal operating conditions.



Event name	Table	Description	Queue	Priority	Fired by	Updated
pa.delete.breakdown	Breakdown [pa_breakdowns]	Created when an breakdown is deleted	pa_clean_scores		PA fire scores clean up event on delete	2016-06-12 23:49:14
pa.delete.indicator	Indicator [pa_indicators]	Created when an indicator is deleted	pa_clean_scores		PA fire scores clean up event on delete	2016-06-12 23:49:14
pa.delete.indicator_breakdown	Indicator Breakdown [pa_indicator_breakdowns]	Created when an indicator breakdown is deleted	pa_clean_scores		PA fire scores clean up event on delete	2016-06-13 00:03:36
pa.job.dc.ended.ok	Job Log [pa_job_logs]	Data collection job ended successfully			Data Collector Job	2014-02-13 03:00:16
pa.job.dc.ended.with.error	Job Log [pa_job_logs]	Data collection job ended with error			Data Collector Job	2014-02-13 03:00:32
pa.job.dc.ended.with.warning	Job Log [pa_job_logs]	Data collection job ended with warning			Data Collector Job	2014-02-13 02:59:15
pa.job.ds.threshold.reached	Threshold definition [threshold_definition]	Fired when a threshold has been reached ...		100	PAThresholdCheckJob	2025-10-14 02:28:51
pa.job.forecast.target.notification	Target [pa_targets]	Fired when forecast meets targets defined			PAForecastJob	2017-01-16 15:21:26
pa.job.indicator.notification		Used to queue email notifications for in...			IndicatorNotificationJob	2013-11-22 01:44:59
pa.job.target.notification	Target [pa_targets]	Fired after comparing with the actual value			PATargetJob	2017-01-15 12:42:31
pa.job.threshold.notification	Threshold [pa_thresholds]	Fired when at least one of all the activ...			PAThresholdJob	2013-12-07 03:13:19
pa.job.threshold.reached	Threshold [pa_thresholds]	Fired when a threshold has been reached			PAThresholdJob	2013-12-07 03:13:43

Sports Hall Booking System Workflow (Step-by-Step)

Step 1: Open Service Portal

- The user opens the **ServiceNow Service Portal**.
- The portal provides access to the Sports Hall Booking System.

Step 2: User Authentication

- The user logs in using valid credentials.
- The system authenticates the user before granting access.

Step 3: Browse Available Sports Halls

- The user views the list of available sports halls.
- Available dates and time slots are displayed in real time.

Step 4: Select Hall, Date, and Time Slot

- The user selects:
 - Sports Hall
 - Booking Date

- Start Time
- End Time

Step 5: Enter Booking Details

- The user fills in the booking form with:
 - Name
 - Contact Number
 - Email Address
 - Purpose of Booking
- The booking details are reviewed before submission.

Step 6: Submit Booking Request

- The user submits the booking request.
- The request is sent to the ServiceNow backend for processing.

Step 7: System Checks Availability

- The system automatically checks the selected hall and time slot.
- It verifies that no conflicting booking already exists.

Step 8: Availability Decision

- **If the slot is available:**
 - The booking process continues.
- **If the slot is unavailable:**
 - The system displays a "**Slot Unavailable**" message.
 - The user is prompted to select another available time slot.

Step 9: Create Booking Record

- If the slot is available, a new booking record is created in the **Sports Hall Bookings** table.
- The booking is assigned an initial status such as **Pending Approval**.

Step 10: Trigger ServiceNow Workflow

- The ServiceNow Flow Designer automatically starts the booking workflow.
- Business Rules and automated processes are executed.

Step 11: Approval Process

- The system checks whether manager approval is required.
- **If approval is required:**
 - The Sports Hall Manager reviews the request.

- The manager either **Approves** or **Rejects** the booking.
- **If approval is not required:**
 - The booking is automatically approved.

Step 12: Send Notification

- The system sends an email notification to the user.
- The notification contains:
 - Booking ID
 - Sports Hall Name
 - Booking Date
 - Time Slot
 - Booking Status (Approved/Rejected)

Step 13: Online Payment (Optional)

- If the selected sports hall requires payment:
 - The user completes the online payment.
 - The payment status is recorded.
- If payment is not applicable, this step is skipped.

Step 14: Update Booking Status

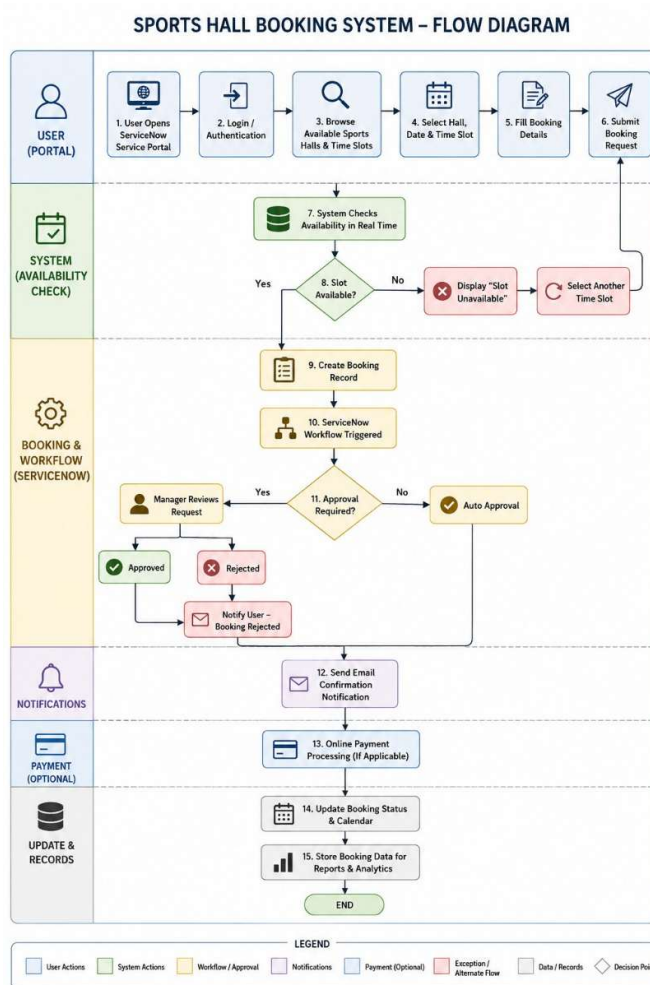
- The booking status is updated automatically.
- The reserved time slot is marked as unavailable.
- The booking calendar is updated accordingly.

Step 15: Store Booking Data

- The final booking details are stored securely in the database.
- The data is used for:
 - Booking history
 - Reports
 - Analytics
 - Facility utilization monitoring
 - Future reference

Step 16: End Process

- The booking process is completed successfully.
- The user can view the confirmed booking through the Service Portal dashboard



Project Demo with Service now Instance ID:

https://dev413692.service-now.com/sp?id=sports_booking

User name: admin

Password: 2UIQf+Bk4bB+